Query Optimisation and Indexing

Query optimisation is central to advanced database systems. The query planner estimates the cost of several plans and picks the cheapest one for each query.

An index is a separate data structure, often a b-tree, that speeds up retrieval at the cost of extra storage and slower writes. A partial index covers only the rows that match a condition.

Use EXPLAIN ANALYZE to display the query plan and the real run time. Concurrency control and transaction processing decide how many queries can safely run at the same time in a distributed database.